Wonjun Jo

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Website | Scholar | Github | LinkedIn

Research Interests

My research focuses on **learning and adaptation** for embodied agents. Building on **self-supervised and multimodal learning**, I study how robots can acquire new capabilities while **preserving and generalizing** what they already know.

My current work explores the adaptation of **pretrained robot policies** and the acquisition of **reusable skills** from demonstrations and interaction. I am particularly interested in how embodied agents can **learn from failures** and transfer their skills to new tasks and environments.

Ongoing Projects

One-shot Teaching for Reusable Robot Skills
Wonjun Jo
In progress

Publications

[1] **Wonjun Jo**, Nam Hyeon-Woo, Yohan Park, Hyunwoo Ha, Tae-Hyun Oh, "When Fine-Tuning Drifts: Mitigating Policy Drift with Base Action-Field Regularization," *Under review*, 2026.

- Mitigating policy drift for robust few-shot fine-tuning of pretrained robot policies.

[2] **Wonjun Jo**, Hyunwoo Ha, Kim Ji-Yeon, Hawook Jeong, Tae-Hyun Oh, "LiDAR-Anchored Collaborative Distillation for Robust 2D Representations," *IEEE Robotics and Automation Letters (**RA-L**)*, 2026. [Paper] [arXiv]

- Improving the robustness and generalization of self-supervised visual representations with LiDAR guidance.

[3] Yohan Park, Hyunwoo Ha, **Wonjun Jo**, Tae-Hyun Oh, "DarkQA: Benchmarking Vision-Language Models on Visual-Primitive Question Answering in Low-Light Indoor Scenes," *IEEE Robotics and Automation Letters (**RA-L**)*, 2026. [Paper] [arXiv]

- Benchmarking the robustness of embodied vision-language models under low-light conditions.

[4] **Wonjun Jo**, Hyunwoo Ha, Kim Ji-Yeon, Hawook Jeong, Tae-Hyun Oh, "Self-Supervised Collaborative Distillation: Enhancing Lighting Robustness and 3D Awareness," Workshop on Wild3D, *IEEE/CVF International Conference on Computer Vision (**ICCV**)*, 2025. [Paper] [OpenReview]

- Improving pretrained visual representations for lighting robustness and 3D-aware generalization.

[5] **Wonjun Jo**, Kwon Byung-Ki, Kim Ji-Yeon, Hawook Jeong, Kyungdon Joo, Tae-Hyun Oh, "The Devil is in the Details: Simple Remedies for Image-to-LiDAR Representation Learning," *Asian Conference on Computer Vision (**ACCV**)*, 2024. [Paper] [arXiv]

- Improving self-supervised image-to-LiDAR representation learning for robust 3D perception.

Patents

Self-Supervised Learning Method, Computing Device, and Computer Program Through Collaborative Distillation Between 2D and 3D Encoders 2024

- Republic of Korea. Application No. 10-2024-0202291; Patent No. 10-2848095 (2025).

Self-Supervised Learning Method, Computing Device, and Computer Program Using Synchronized Sensor Data and Unsynchronized Sensor Data 2023

- Republic of Korea. 10-2023-0146900

Projects

Modular and Agentic Physical AI (KIST) Jul. 2026 – Present

- Modular robot architectures linking perception, planning, and skill execution through agentic programming.
- Robust vision and tactile feedback for dexterous manipulation with high-DoF hands.
- Cooking as a testbed for long-horizon planning, task-state tracking, and precise bimanual manipulation.

Self-Supervised Learning Method Utilizing Camera-LiDAR Correlation for Autonomous Driving Jan. 2023 – Dec. 2024

- Proposed a novel self-supervised learning method for improving 2D representation using LiDAR.
- Proposed remedies for improving self-supervised image-to-LiDAR distillation method.
- Funded by the RideFlux Inc., South Korea.

Deformable Object Recognition Technology Research Mar. 2022 – Dec. 2022

- Developed deformable object recognition algorithm on multi-spectral images.
- Developed motion vector estimation algorithm for deformable objects using LiDAR.
- Funded by the Agency for Defense Development (ADD), South Korea.

Academic Services

Journal Reviewer

- The Visual Computer: International Journal of Computer Graphics (**TVCJ**), 2023

Conference Reviewer

- International Conference on Robotics and Automation (**ICRA**), 2023

Education

POSTECH Pohang, South Korea
Joint M.S & Ph.D. in Electrical Engineering (Advisor: Tae-Hyun Oh) *Mar. 2022 –*

Inha University Incheon, South Korea
B.S. in Computer Engineering *Mar. 2018 – Feb. 2022*

Experience

KIST Physical AI Research Scientist Internship Seoul, South Korea
Korea Institute of Science and Technology (Advisor: In So Kweon) *Jul. 2026 – Present*

POSTECH → KAIST Pohang → Daejeon, Korea
Research Assistant (Advisor: Prof. Tae-Hyun Oh) *Feb. 2022 – Present*

POSTECH Student Internship Pohang, South Korea
Pohang University of Science and Technology (Advisor: Tae-Hyun Oh) *Jun. 2021 – Aug. 2021*

Extra

Deep Learning Seminar

- POSTECH AMILab seminar [YouTube]
- POSTECH EE-group deep learning seminar
- Attending POSTECH PAIR (CVLab, CGLab, MLLab, AMILab) seminar

Teaching Experiences

- [EECE454] Introduction to Machine Learning, POSTECH